



BT605 FINAL TERM Solved Questions

semester Fall 2021

Ethical Issues Related to therapeutic cloning:-

moral status of the embryo ---- destruction • patient has the right to live • morally right for in vitro fertilization but morally wrong to save a child's life • potential donor exploitation slippery slope ----- reproductive cloning different from therapeutic cloning • no access and benefit to poor communities • raises issues of social justice and healthcare disparities

Information Security:-

Introduction: • defending information from unauthorized access • use, disrupt, disclose, recorded, inspected, modified • software attacks viruses, worms, trojan horses

Elements: • confidentiality "property" not disclosed to unauthorized persons • integrity-maintaining and assuring the accuracy and completeness of data • availability

Duties of Fire Warden:-

• a person employed to prevent / extinguish fire • important risk measures • raise awareness among staff • how to respond in emergency ensure evacuation • helping-wheelchair • switch off electrical appliances • close the doors to isolate fire • guide everyone to assemble area ensure evacuation •

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Personal Protective Equipment:-

• to protect hands from hazardous materials • glove selection---- risk assessment • latex gloves- available • wear outside-lab • wear two pairs ----- required • gowns/coats/smocks/ uniforms designated for lab • prevent personal clothing • remove-leaving for non-laboratory areas • deposit for laundry/ should not be taken home

Eye and face protection: • goggles, mask, face shield, splash guard • contact lenses • dispose of/ decontaminate • used in rooms ----- infected animals

Respirators: • inhalation of aerosols • full/half face • interchangeable filters • shouldn't worn outside lab • disposable-respirator

High Containment Labs:-

Tsunami - blame government-lack of preparation • Tsunami in labs • biosafety and biosecurity-Africa and south Asian countries

Control: • responsibility of scientific community • potential for both accidental and malicious breeches • what is safe to be used?

High containment facility design: • double-door entries • directional/negative pressure air flow • single-pass air • air changes per hour • multiple safety measures

Still danger: • global warming • high energy cost • bioterrorism • green technology approach • tax incentives, carbon tax, rising fossil fuel

Nations with limited resources: • elect officers /ministers • formulate national plan • allocate budget • local economy - services, reagents and equipments • technical information

Global control: • organizations • legislations • guidelines

Fire Exit:-

Rules for fighting fire: • fire is small and contained • You are safe from toxic smoke • means of escape • your instincts tell you it okay

kind of emergency exit mounted to the outside of a building • faster evacuation • alternative routes when regular exit is blocked

History: • 1883-England-180 children died • 1911- America- 146 factory worker died • 9/11- exit doors were locked • all buildings have well - marked emergency exits

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Signage: • “EXIT” • running green man • Introduced in 2003 by ISO 7010

Lab Biosecurity:-

Objectives: • this supports lab safety agenda to prevent diseases • ensure containment of infectious materials • maintain citizen confidence of bioscience research community • transparency to investors in the industries • protect valuable research and commercial assets • reduce the risk of crime and bioterrorism

Japan Biosecurity Legislations:-

Introduction: • Japan ministry of health, labor and welfare • two pillars of biosecurity • surveillance of infection and infectious agents • regulations of pathogen handling

Duties: • screening of foods, human, vectors at the point of entry • Japan ministry of agriculture, forest and fisheries • health issues-animals and plants • bioweapon prohibition laws

Noise Risk and Control :-

Risk: • noise measurement surveys • specify areas • lab equipment(laser) • animal house

Control: • barriers around noisy equipments • barrier between noisy areas and work areas • hearing conservation program • medical monitoring program

Classification of Pathogens:-

• **WHO risk group 1:** microbes unlikely to cause disease

WHO risk group 2: microbes causing diseases-unlikely to be serious

WHO risk group 3: pathogens causing serious disease

WHO risk group 4: pathogens causing serious disease ---- transmission-----no effective treatment or preventive measures

Elements of personnel security:-

• personnel screening • badges • visitors control • training • all positions must be defined and trained • security issues must be addressed • divide responsibilities • security officers---- personnel security policies

Route of Exposure of Hazardous chemicals:-

• inhalation • contact • ingestion • needle sticks • broken skin

Biological security kit:-

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• household bleach • 70% alcohol • spray bottle • sterilization bags • absorbent paper towels • yellow trash bags • disposable gloves • metallic tongs • surgical masks • safety goggles, shoe covers and face mask • spill control and cleanup procedures

Duties of Biosafety Officer:-

• apply National or International rules, regulations/guidelines • assist lab in developing standard safety procedures • knowledge of lab, clinical practices • knowledge of devices/engineering principles • maintenance facilities • communicate with administrative, technical and support personnel

Fire Hazard:-

Introduction: • close cooperation between safety officer and fire prevention officer • immediate action in case of fire • determine-it is best to contain or extinguish fire

Reasons: • electric circuit overloading • poor electrical maintenance • long electrical leads • equipment unnecessarily switched on • open flames • equipments not designed for lab environment • improper ventilation • mishandling of the chemicals

Fire-fighting equipments: • inspected, maintained shelf life • near doors/ corridors • hoses, buckets and fire extinguishers • fire warnings/ instructions • escape routes/ assembly point

Types: • water: wood, fabric, paper • carbon dioxide gases: flammable liquids and gases/electrical fires • dry powder: flammable liquids and alkali metals • foam: flammable liquids

Low risk TB Labs:-

• minimum biosafety requirements • direct sputum-smear microscopy • preparation of specimen-automated nucleic acid purification assay

Factors increasing the risk of infection: • improper bench spaces • specimen container may leak • specimen manipulated carelessly • specimen must be shaken vigorously • ventilation may be poor

Risk Reduction by Risk Management:-

• structured approach • occur in steps • identify, assess and control risk • processes in place to minimize and monitor risk • control adverse risk consequences materialize

Steps: • early consultation identify needs and cost • deferring irreversible decisions-need more time to achieve objectives • pilot studies - more information about risk • design flexibility modify-future needs precautionary actions • transferring risk to private sector ----- insurance • less use of leading edge technology • undertaking site investigation staging the project review at different steps • abandoning the project - too risky

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Aarhus Regulation:-

- grants public rights • imposes obligations • community/institution access environmental information

Risk of Ionizing Radiation:-

- somatic effects: radiation induced cancers • hereditary effects of radiation exposure to the gonads

UK-cost benefit analysis:-

- analysis of procedure and experiments • number and type of animals used • must be weighed against the potential benefits of the project.

First Aid:-

- Definition:** • medical treatment--- time and place of an accident • approved method • treat the casualty • before doctor's care for treatment

- Contains??:** • first aid box • protective clothing • safety equipment • eye irrigation

- First aid box:** • made up of material keep the content dust and damp free • white cross with green box • prominent position • easily recognizable • first aid manual • bandages • sterile dressings • safety pins

- Protective equipment:** • mouth piece for mouth-to-mouth resuscitation • gloves-protections • clean up kit for blood spills

Immunization of Staff:-

- Introduction:** • discuss with workers • vaccines • therapeutic drugs after exposure

Genetic Screening:-

- Definition:** • systemic search for person with specific genotype • individual or group show a risk of disease • genetic testing— specific or multiple gene interaction

- DNA:** • only requirement • common thread of life • provide life its blueprint for building, replicating and surviving • condenses to form chromosomes • allelic pairs make up genes

- Ethics:** • religious groups • expensive/emotional distress • done for common diseases/part of medical record??? • deny employment, social services and insurance benefits

Electrical Hazards:-

- Introduction:** • electrical appliances and equipments ----- tested and inspected • electric- circuit protect wiring from being overloaded with electric current • earth-fault interrupters: protect people from electric shock

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Safety: • National electrical standard and safety codes • earth / grounding systems with threeprong plugs

Patient Rights:-

• give correct/complete information • ask questions • cooperate with your caregivers • accept health consequences • respect and privacy • quality care • information and communication • make decisions

BSL-3:-

• applicable to diagnostic / clinical/ research/ production/ teaching facilities • potentially lethal disease through the inhalation route • all procedures must be performed in a biosafety cabinet

Equipments: • vacuum lines must be protected with HEPA filters • ducted air ventilation system • HEPA filter exhaust air • BSL-3 facility design, operational/parameter /procedures must be documented

BSL-4:-

• dangerous exotic agents • aerosol transmitted lab infections • no vaccine/treatment • unknown risk of transmission

Types: • A cabinet laboratory manipulation of agents in BSC • A suit laboratory personnel must wear positive pressure air protective suit

Biosafety measures for TB lab:-

Concepts: • lab access • responsibilities of lab manager • personnel protective equipment • work areas • equipment • waste handling • incineration • autoclaving • disinfection

Types of Hazardous Chemicals:-

• toxic chemicals • explosive chemicals • compressed and liquefied gases

Explosive chemicals: • Azides shouldn't react with metals • Ether that have aged, dry crystals-unstable • Perchloric acid, not dry on wood or fabric • Picric acid, picrates explode by heat

Gases: • chained with the wall • stored in a separate room • away from heat/ open flames/radiators electrical appliances • must not be incinerated

Effect of Global Warming:-

• earth climate zone is shifting • Polar ice start melting • Sea level increases • metabolic rate of methane producing bacteria increases • Species may extinct

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Biosafety Cabinet:-

- Class I, II, III BSC • air intake velocity • amount of air circulated • exhaust system • pressure system

Confidential Information:-

- privileged or property information • privileged information shared among few people for further processing • unauthorized people shouldn't take advantage • Information of the applicant • protected with article 21 of the Cartagena protocol • set forth in the biosafety guidelines

How to respond a Disaster?

Definition: • event occurring suddenly-causing loss of life damage or hardship • sudden overwhelming and unforeseen event

Disaster response: • second phase of disaster management cycle • warning/evacuation/ search/rescue • immediate/continuing assistance • assessing damage/ restoration infrastructure

Types: • at household level • at community level

Aims: • assistance to maintain life • improve health • support the morale affected population • limited aid

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Separate WhatsApp groups for girls and boys

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